

GEOGRAPHY 05 | PREMIUM SOLVED WORKBOOK

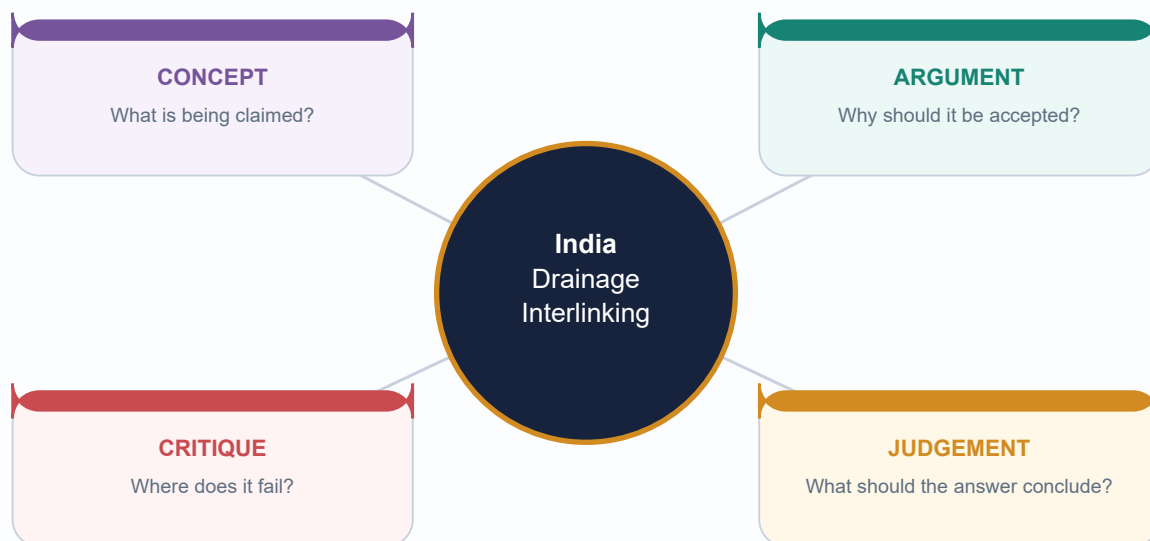
India Drainage and Interlinking - Solved Workbook

Official papers checked locally | Geography | Prelims + GS-I

Official 2024 keys separated from inferred/preliminary-key answers

Original MCQ answers rotate strictly A -> B -> C -> D; PYQ options are retained verbatim.

Every Mains model follows claim -> named evidence -> analysis -> qualification -> verdict.



Facts from official sources | Inferences clearly marked | No fabricated data

HIGH UPSC RELEVANCE

HIGH

PYQ evidence register: ownership and answer-key status

GS-I | Prelims
Geography

INTRO & ORIGIN

[FACT] Direct Topic 05 ownership is preserved; adjacent water questions are identified rather than silently absorbed.

STATIC THEORY

- **Direct owner:** 2020 GS-I Q14 interlinking; 2021 Prelims Q53 Indus direct junction and Q55 Eastern-Ghats sources; 2022 Q24 Gandikota/Pennar; 2024 Q9 waterfalls; 2026 Q11 drainage history and Q25 antecedent river.

- **Relevant map extensions:** 2019 Prelims Q23 famous-place/rivers; 2023 Prelims Q1 rivers/lakes; 2024 Prelims Q6 Ganga tributary order. These remain linked to their primary geography owners but help India-drainage mapwork.

- **Adjacent owner, included for boundary awareness:** 2018 Prelims Q81 heavy sand mining in riverbeds is Environment-and-Ecology-owned; its solution is included because it tests a river-system consequence, not because this package claims regulatory/pollution depth.

- **Key rule:** 2024 Set-A answer key is official and held locally. 2018-2023 keys are unavailable locally. 2026 key is provisional locally. All non-official answers below are explicitly labelled INFERRED ANSWER - NOT OFFICIALLY VERIFIED.

MAINS ANGLE

For Mains, every answer below uses claim -> named evidence -> analysis -> qualification and ends with Why this earns marks.

STUDY LINK

Central routing ledgers and official-paper files listed in source register.

HIGH

Solved Mains PYQ - 2020 GS-I Q14

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): The interlinking of rivers can provide viable solutions to the multi-dimensional inter-related problems of droughts, floods and interrupted navigation. Critically examine. (Answer in 250 words)

STATIC THEORY

- **Claim:** Interlinking can be viable in selected, rigorously assessed corridors, but it cannot be presumed to solve drought, floods and navigation together because each depends on different hydrological and institutional conditions.
- **Drought - named evidence -> analysis:** NWDA's current Ken-Betwa material describes a Ken-to-Betwa transfer under implementation. Storage/conveyance can alter seasonal water availability for a defined recipient. This does not erase rainfall deficit, groundwater depletion, distribution losses or crop-demand mismatch; donor balance must reserve existing/downstream/ecological needs.
- **Flood - named evidence -> analysis:** A reservoir can attenuate a specified peak only if designed/operated with flood space. In an extreme or simultaneous basin event, transfer capacity, sedimentation, releases and floodplain exposure constrain the effect. Floodplain restoration/zoning and forecasts therefore complement, not follow from, a link.
- **Navigation - named evidence -> analysis:** A regulated route may help a particular corridor, but depth, locks, terminals, sediment management, seasonal flow and ecological safeguards decide whether navigation is viable. A canal is not automatically a navigation network.
- **Qualification:** Panna landscape concerns, downstream-flow/sediment effects, local social impacts and MP-UP-Union coordination show that ecological/social/federal feasibility is part of hydraulic feasibility. NWDA April 2026 status evidence proves activity, not delivered outcomes.
- **Verdict:** Use interlinking selectively after seasonal basin accounting and alongside watershed/recharge, demand/crop management, wastewater reuse, reservoir operations and floodplain restoration.
- **Why this earns marks:** It answers all three limbs, uses dated official case evidence accurately, explains mechanism and limitation for each, and gives a conditional critical verdict rather than a slogan.

MAINS ANGLE

Compact sketch: donor basin -> storage/conveyance -> recipient, with four gates: seasonal balance, e-flow/sediment, demand/operations, social/federal consent.

STUDY LINK

Source: [books/more_previous_papers/Gen_St_P1.pdf](#); local Mains routing ledger.

HIGH

Solved Prelims PYQ - 2018 Q81 (adjacent Environment owner)

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Which of the following are the possible consequences of heavy sand mining in riverbeds?

1. Decreased salinity in the river 2. Pollution of groundwater 3. Lowering of the water-table. Select the correct answer: A. 1 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B, 2 and 3 only. Confidence: high.**

- **Elimination:** Heavy sand removal can disturb riverbed/aquifer connection and lower local water tables; pollution risk can arise through disturbance/associated activity. 'Decreased salinity' is not a general consequence of sand mining and cannot be safely assumed.

- **Ownership qualification:** The local ledger routes this to Environment-and-Ecology. This package uses it only to reinforce riverbed-groundwater/floodplain connectivity, not to replace water-pollution regulation depth.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/QP-CSP-18-GS-I-C.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2019 Q23 famous place and river pairs

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Consider the pairs: 1. Pandharpur - Chandrabhaga 2. Tiruchirappalli - Cauvery 3. Hampi - Malaprabha. Which pairs are correctly matched? A. 1 and 2 only B. 2 and 3 only C. 1 and 3 only D. 1, 2 and 3.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A, 1 and 2 only. Confidence: high.**

- **Elimination:** Pandharpur is associated with Chandrabhaga (the local name of the Bhima bend); Tiruchirappalli is on the Cauvery. Hampi is associated with the Tungabhadra, not the Malaprabha.

- **Map lesson:** Pair a cultural/place cue with the river system, then test the distractor by locating it in the correct Peninsular sub-basin.

MAINS ANGLE

STUDY LINK

Official paper:
[books/more_previous_papers/csp-p1.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2019 Q38 glacier and river source pairs (map extension)

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Consider: 1. Bandarpunch - Yamuna 2. Bara Shigri - Chenab 3. Milam - Mandakini 4. Siachen - Nubra 5. Zemu - Manas. Which pairs are correctly matched? A. 1, 2 and 4 B. 1, 3 and 4 C. 2 and 5 D. 3 only.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A, 1, 2 and 4. Confidence: high.**

- **Elimination:** Bandarpunch-Yamuna, Bara Shigri-Chenab and Siachen-Nubra are correct. Milam is associated with the Gori Ganga system rather than Mandakini; Zemu is associated with Teesta, not Manas.

- **Ownership qualification:** The primary owner is glacial geography (Topic 06). It is retained as a source-map extension because headwaters matter in Indian drainage mapping.

MAINS ANGLE

STUDY LINK

Official paper:
[books/more_previous_papers/csp-p1.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2020 Q68 world-river mouths (map extension)

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Consider: 1. Mekong - Andaman Sea 2. Thames - Irish Sea 3. Volga - Caspian Sea 4. Zambezi - Indian Ocean. Which pairs are correctly matched? A. 1 and 2 only B. 3 only C. 3 and 4 D. 1, 2 and 4 only.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: C, 3 and 4. Confidence: high.**

- **Elimination:** Mekong reaches the South China Sea, not the Andaman Sea; Thames reaches the North Sea, not the Irish Sea. Volga reaches the Caspian Sea and Zambezi reaches the Indian Ocean.

- **Ownership qualification:** This is World-Regional Geography's primary map route. It is included to train the same source-course-mouth discipline used for India, not to broaden this package into a world-river catalogue.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/CSP_2020_GS_Paper-1.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2021 Q53 Indus direct junction

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): With reference to the Indus river system, of the following four rivers, three of them pour into one of them which joins the Indus direct. Which one is such a river? A. Chenab B. Jhelum C. Ravi D. Sutlej.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: A, Chenab. Confidence: high.**

- **Elimination:** Jhelum and Ravi join Chenab; Beas joins Sutlej and the combined lower Punjab tributary relation reaches Chenab/Panjinad before the Indus. Thus Chenab is the direct trunk asked by the question's network logic.

- **Map lesson:** Do not learn the five names as a flat list; draw converging arrows toward the Indus and identify the final trunk.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/QP-CSP-21-GeneralStudiesPaper-I-121021.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2021 Q55 Eastern Ghats sources

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Consider: 1. Brahmani 2. Nagavali 3. Subarnarekha 4. Vamsadhara. Which rise from the Eastern Ghats? A. 1 and 2 B. 2 and 4 C. 3 and 4 D. 1 and 3.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B, 2 and 4 only. Confidence: high.**

- **Elimination:** Nagavali and Vamsadhara are Eastern-Ghats source cues. Brahmani and Subarnarekha do not fit the stated source condition.

- **Map lesson:** Source terrain, flow direction and mouth are separate tests; do not infer source from an east-coast mouth.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/QP-CSP-21-GeneralStudiesPaper-I-121021.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2022 Q24 Gandikota canyon

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Gandikota canyon of South India was created by which river? A. Cauvery B. Manjira C. Pennar D. Tungabhadra.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: C, Pennar. Confidence: high.**

- **Elimination:** Gandikota is cut by the Pennar. The question tests a river-landform-location relationship, not a generic claim that every canyon is Himalayan or that every Peninsular river is low-energy.

- **Map lesson:** Link landform -> river -> regional setting; revisit profile breaks, resistant rock and local incision.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/GENERAL STUDIES PAPER I.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQ - 2023 Q1 river and lake relations

GS-I | Prelims
Geography

INTRO & ORIGIN

Question (official local paper): Consider: 1. Jhelum River passes through Wular Lake. 2. Krishna River directly feeds Kolleru Lake. 3. Meandering of Gandak River formed Kanwar Lake. How many statements are correct? A. Only one B. Only two C. All three D. None.

STATIC THEORY

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: B, only two. Confidence: high.**

- **Elimination:** Jhelum-Wular is correct. Kolleru is directly fed by Budameru and Tammileru rather than Krishna. Kanwar/Kabar is an oxbow lake associated with Gandak meandering.

- **Ownership qualification:** The primary owner is Topic 09 Lakes; retain it here as a river-process/map extension, not as a claim to wetland-conservation depth.

MAINS ANGLE

STUDY LINK

Official paper: [books/more_previous_papers/QP_CS_Pre_Exam_2023_280523.pdf](#); key unavailable locally.

HIGH

Solved Prelims PYQs - 2024 Q6/Q9 official-key set

GS-I | Prelims
Geography

INTRO & ORIGIN

[FACT] The 2024 official Set-A answer key is held locally, unlike the older questions above.

STATIC THEORY

- **Q6 question:** With reference to Himalayan rivers joining the Ganga downstream of Prayagraj from west to east, which sequence is correct? A. Ghaghara-Gomati-Gandak-Kosi B. Gomati-Ghaghara-Gandak-Kosi C. Ghaghara-Gomati-Kosi-Gandak D. Gomati-Ghaghara-Kosi-Gandak.

- **OFFICIAL ANSWER: B.** Gomati joins before Ghaghara in the specified west-east order, followed by Gandak and Kosi. Map the stated reach; the answer is not a generic tributary list.

- **Q9 question:** Consider: 1. Dhuandhar - Nagpur 2. Hundru - Chota Nagpur - Subarnarekha 3. Gersoppa - Western Ghats - Netravati. In how many rows is the information correctly matched? A. Only one B. Only two C. All three D. None.

- **OFFICIAL ANSWER: A, only one.** Hundru/Chota-Nagpur/Subarnarekha is correct. Dhuandhar is on Narmada near Jabalpur; Gersoppa/Jog is on Sharavathi, not Netravati.

MAINS ANGLE

STUDY LINK

Official paper/key: [books/prelima_question_per_answers/2024-GS1-Set A.pdf](#) and [Ans-2024-GS1.pdf](#).

HIGH

Solved Prelims PYQs - 2026 Q11/Q25 provisional-key set

GS-I | Prelims
Geography

INTRO & ORIGIN

[FACT] The locally held 2026 Set-A key is labelled provisional. The answers below are therefore inferential, not official.

STATIC THEORY

- **Q11 question:** Pleistocene assertion: either Yamuna once flowed into Indus, or Sutlej flowed into Yamuna and a major tributary shifted between Ganga and Indus. Bases listed: 1. Nadi-Sukta of the Rigveda 2. Robert Bruce Foote's explorations of Sutlej/Yamuna 3. same dolphin species in Indus and Ganga systems. Select the basis/bases using the printed code.

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: the option containing 1, 2 and 3. Confidence: moderate.**

Textual relative-river evidence, geomorphic exploration and shared river-dolphin biogeography are presented as converging evidence routes; they do not yield a simple directly observed Pleistocene map. The uncertainty is why the answer is not presented as an official key.

- **Q25 question:** Identify the river: 1. antecedent drainage 2. flows through three countries 3. Tibetan Plateau origin and important for irrigation 4. does not form distributaries. A. Brahmaputra B. Indus C. Sutlej D. Teesta.

- **INFERRED ANSWER - NOT OFFICIALLY VERIFIED: C, Sutlej. Confidence: high.** Sutlej fits the transboundary/Tibetan/irrigation and non-distributary contrast; Indus is undermined by its delta/distributary condition in the stem's comparison.

MAINS ANGLE

STUDY LINK

Official paper: [books/prelima_question_paper_answers/2026-GS1-Set A.pdf](#); provisional key held locally.

HIGH

Hard MCQs - questions 1-8 (attempt before solutions)

GS-I | Prelims
Geography

INTRO & ORIGIN

Original questions only. No answer or hint appears in this block. Correct options across original MCQs rotate strictly A -> B -> C -> D.

STATIC THEORY

- **Q1.** A drainage basin is best defined as:

A. Land draining to a stated outlet | B. A single deepest channel reach | C. Any surface-water storage | D. A river's entire wetted channel

- **Q2.** Which statement correctly distinguishes competence from capacity?

A. Competence is total load; capacity is particle size | B. Competence is largest movable particle; capacity is total load | C. Both describe basin area | D. Both describe water quality

- **Q3.** Which is the most defensible inference from a knickpoint?

A. It proves a river is permanently youthful | B. It proves a glacier cut the valley | C. It is a profile break that may reflect local base level, structure or rejuvenation | D. It proves the river cannot deposit downstream

- **Q4.** A rectangular drainage pattern most strongly indicates:

A. Uniform unjointed rock | B. An internal basin | C. A volcanic cone | D. Joint/fault control

- **Q5.** An antecedent river:

A. Predates uplift and maintains course by incision | B. Develops only after a cover is removed | C. Is a sudden floodplain avulsion | D. Must be a glacier-fed river

- **Q6.** River capture is most directly caused by:

A. Tidal scour at a mouth | B. Headward erosion and divide migration | C. Chemical solution at a spring | D. A delta distributary bifurcation

- **Q7.** Which statement about Peninsular drainage is correct?

A. Every river is seasonal | B. Every major river flows east | C. Most large systems flow east, but Narmada and Tapi are major west-flowing structural exceptions | D. Only rivers from Eastern Ghats flow east

- **Q8.** Which river-mouth pairing is correct?

A. Godavari - Gulf of Khambhat estuary | B. Tapi - Bay of Bengal delta | C. Luni - normal perennial open-sea mouth | D. Narmada - Gulf of Khambhat estuary

MAINS ANGLE

Write a reason for eliminating each distractor before checking the next solution block.

STUDY LINK

See the immediately following solution card.

HIGH**Hard MCQs - questions 9-16 (attempt before solutions)**

GS-I | Prelims
Geography

INTRO & ORIGIN

Original questions only. No answer or hint appears in this block. Correct options across original MCQs rotate strictly A -> B -> C -> D.

STATIC THEORY

- **Q9.** A delta is most likely to persist where:

A. Sediment delivery can keep pace with removal/subsidence over time | B. Any large river reaches the sea | C. A river has no tributaries | D. Tides are always absent

- **Q10.** A natural floodplain is best understood as:

A. A zone with no hydrological function | B. A depositional/inundation surface with storage, recharge and habitat functions | C. An artificial reservoir | D. A legal category identical to encroachment

- **Q11.** Which claim about inter-basin transfer is most accurate?

A. An annual average alone establishes surplus | B. Construction proves benefits have been delivered | C. A donor balance must account for season, storage, existing use and ecological/downstream needs | D. A canal automatically moderates every flood

- **Q12.** For navigation, a transfer proposal must additionally establish:

A. Only a donor basin name | B. Only an irrigation command | C. Only an annual rainfall average | D. Reliable depth, works/terminals, sediment management and ecological safeguards

- **Q13.** The current official Ken-Betwa direction is:

A. Ken basin -> Betwa basin | B. Betwa basin -> Ken basin | C. Narmada -> Betwa basin | D. Ganga -> Ken basin

- **Q14.** The safest interpretation of the April 2026 NWDA KBLP update is:

A. Final benefits are verified | B. Implementation/progress activity is reported, not final outcome delivery | C. The project is only a hypothetical map line | D. No institution is involved

- **Q15.** A sound Panna landscape appraisal should include:

A. Only a construction schedule | B. Only a projected water benefit | C. Landscape/connectivity, downstream flow, social effects, mitigation and monitoring | D. Only a wildlife label

- **Q16.** Which is most clearly a complement to, rather than a synonym for, inter-basin transfer?

A. Treating annual average as fixed surplus | B. Ignoring crop demand | C. Equating channelisation with restoration | D. Watershed/recharge, demand management, wastewater reuse and reservoir operations

MAINS ANGLE

Write a reason for eliminating each distractor before checking the next solution block.

STUDY LINK

See the immediately following solution card.

HIGH**Hard MCQs - questions 17-24 (attempt before solutions)**

GS-I | Prelims
Geography

INTRO & ORIGIN

Original questions only. No answer or hint appears in this block. Correct options across original MCQs rotate strictly A -> B -> C -> D.

STATIC THEORY

- **Q17.** Which answer structure best fits 'Critically examine interlinking'?

A. Potential mechanism -> named case -> limits/safeguards -> conditional verdict | B. River list -> generic conclusion | C. Only project criticism | D. Only project benefits

- **Q18.** The named Ganga convention most safely states that:

A. Ganga begins only at Gomukh | B. Ganga is named at Devprayag after Bhagirathi-Alaknanda confluence | C. Alaknanda is a distributary | D. Devprayag lies in the delta

- **Q19.** Which pairing is correct?

A. Gersoppa/Jog - Netravati | B. Dhuandhar - Godavari | C. Hundru - Subarnarekha | D. Gandikota - Cauvery

- **Q20.** Which statement best distinguishes river capture from an oxbow?

A. Both are delta processes | B. Both are structural troughs | C. Both require a tidal estuary | D. Capture diverts a headwater by headward erosion; an oxbow follows meander cut-off

- **Q21. [REMEDIAL]** Remedial: which evidence most strongly supports an antecedent interpretation?

A. A river cuts a transverse gorge while maintaining a course through uplifted relief | B. A river has a wide delta | C. A lake sits in an internal basin | D. A channel takes right-angle bends

- **Q22. [REMEDIAL]** Remedial: which statement about Himalayan flow is safest?

A. All flow is from glaciers | B. Most large systems are perennial, but rain/snow/ice/groundwater contributions and discharge vary seasonally | C. Every reach has equal discharge | D. No Peninsular river is perennial

- **Q23. [REMEDIAL]** Remedial: a donor basin is called 'surplus' only after:

A. A project brochure uses the word | B. It has more annual rain than a recipient | C. Seasonal balance, storage, existing/downstream/e-flow needs and uncertainty are assessed | D. A canal is commissioned

- **Q24. [REMEDIAL]** Remedial: which project statement is factually disciplined?

A. Daudhan progress proves all benefits are complete | B. Ken-Betwa reverses Betwa into Ken | C. Panna effects can be dismissed after one mitigation plan | D. An implementation update must be separated from commissioning and observed outcome

MAINS ANGLE

Write a reason for eliminating each distractor before checking the next solution block.

STUDY LINK

See the immediately following solution card.

HIGH

Hard MCQs - solutions 1-8

GS-I | Prelims
Geography

INTRO & ORIGIN

Validated original-answer sequence: Q1 A, Q2 B, Q3 C, Q4 D, Q5 A, Q6 B, Q7 C, Q8 D.

STATIC THEORY

- **Q1: A.** A basin is the outlet-linked contributing area; the channel is only one pathway within it.
- **Q2: B.** Competence concerns maximum particle size; capacity concerns the total sediment quantity a flow can carry.
- **Q3: C.** Knickpoints mark a profile discontinuity; their cause must be investigated rather than assumed.
- **Q4: D.** Right-angle bends commonly reveal structural discontinuities such as joints and faults.
- **Q5: A.** Antecedence is a chronology relative to uplift, not a discharge regime.
- **Q6: B.** A more vigorous stream erodes headward and diverts a neighbouring headwater.
- **Q7: C.** The broad eastward tilt has important exceptions; regime is also not uniform.
- **Q8: D.** Narmada and Tapi are west-coast/gulf estuarine examples; Luni trends toward the Rann.

MUST-KNOW FACTS

1. Q1-20 span definitions, process, India systems, mouth/floodplain logic, interlinking and governance.
2. Q21-24 are remedial: anomalous drainage, regimes, surplus accounting and project-stage discipline.

UPSC TRAPS

WRONG: Remembering a letter without mechanism.

CORRECT: Name the clue that makes the other options fail.

MAINS ANGLE

If two questions from one family are wrong, redraw the relevant visual and retry without notes.

STUDY LINK

Return to the corresponding teaching section before another attempt.

HIGH

Hard MCQs - solutions 9-16

GS-I | Prelims
Geography

INTRO & ORIGIN

Validated original-answer sequence: Q9 A, Q10 B, Q11 C, Q12 D, Q13 A, Q14 B, Q15 C, Q16 D.

STATIC THEORY

- **Q9: A.** Delta form is a sediment-budget outcome, not a simple river-size rule.
- **Q10: B.** Floodplains are part of the river system; exposure and encroachment are separate questions.
- **Q11: C.** Transfer feasibility requires joint, time-sensitive water accounting and non-hydraulic tests.
- **Q12: D.** Navigation has operational, infrastructure and ecological prerequisites beyond water transfer.
- **Q13: A.** NWDA's current project material states diversion from Ken to the water-deficit Betwa basin.
- **Q14: B.** Status evidence must not be inflated into irrigation, power or social outcome evidence.
- **Q15: C.** A project trade-off is ecological, hydrological, social and temporal.
- **Q16: D.** Different interventions address different points in the water-security chain.

MUST-KNOW FACTS

1. Q1-20 span definitions, process, India systems, mouth/floodplain logic, interlinking and governance.
2. Q21-24 are remedial: anomalous drainage, regimes, surplus accounting and project-stage discipline.

UPSC TRAPS

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CORRECT: Name the clue that makes the other options fail.

MAINS ANGLE

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STUDY LINK

Return to the corresponding teaching section before another attempt.

HIGH

Hard MCQs - solutions 17-24

GS-I | Prelims
Geography

INTRO & ORIGIN

Validated original-answer sequence: Q17 A, Q18 B, Q19 C, Q20 D, Q21 A, Q22 B, Q23 C, Q24 D.

STATIC THEORY

- **Q17: A.** Critical examination tests viability rather than preaching approval or rejection.
- **Q18: B.** State the convention: a headstream can differ from the named-river formation.
- **Q19: C.** Hundru is on the Subarnarekha; the other pairs misuse the river-landform relation.
- **Q20: D.** They operate at different scales and by different mechanisms.
- **Q21: A.** Antecedence concerns maintained course relative to uplift, usually recorded by transverse incision.
- **Q22: B.** Perennial is not uniform; multiple sources contribute in changing proportions.
- **Q23: C.** Surplus is an accounting conclusion with climate and ecological conditions, not a fixed label.
- **Q24: D.** Status-stage precision prevents a construction claim becoming a delivery claim.

MUST-KNOW FACTS

1. Q1-20 span definitions, process, India systems, mouth/floodplain logic, interlinking and governance.
2. Q21-24 are remedial: anomalous drainage, regimes, surplus accounting and project-stage discipline.

UPSC TRAPS

WRONG: Remembering a letter without mechanism.

CORRECT: Name the clue that makes the other options fail.

MAINS ANGLE

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STUDY LINK

Return to the corresponding teaching section before another attempt.

HIGH

Original 10-mark practice - fluvial adjustment

GS-I | Prelims
Geography

INTRO & ORIGIN

Question: Explain why a river can erode, transport and deposit within a short distance without fitting a rigid upper-middle-lower-course sequence. Answer in 150 words.

STATIC THEORY

- **Claim:** River work responds to local energy relative to load, so the same river can switch dominant behaviour over short distances; the course model is a tendency, not a time law.
- **Named evidence/example:** Steep Himalayan headwaters commonly incise gorges/rapids. Gandikota canyon cut by the Pennar shows that local structure/base-level conditions can create strong incision in Peninsular terrain.
- **Analysis:** Tributaries alter discharge/load; resistant beds create waterfalls/knickpoints; a widening low-gradient alluvial reach permits lateral erosion, bars, floodplains and oxbows. The Ganga-Brahmaputra-Meghna system shows prolonged transport/deposition toward a deltaic coast.
- **Qualification:** Uplift, dams/backwater, floods and rock resistance repeatedly interrupt the profile. Erosion, transport and deposition coexist in one reach.
- **Verdict:** Diagnose local gradient, load, discharge and boundary resistance before assigning a landform to a 'course'.
- **Why this earns marks:** It directly explains the apparent contradiction, uses two named Indian examples, connects evidence to process and corrects the rigid-stage misconception.

MAINS ANGLE

Sketch: concave profile + knickpoint + floodplain; label the process rather than every landform.

STUDY LINK

GC Leong long-profile evidence; 2022 Gandikota PYQ.

HIGH**Original 15-mark practice - Indian drainage comparison**

GS-I | Prelims
Geography

INTRO & ORIGIN

Question: Compare Himalayan and Peninsular drainage systems. How do temporal regime and structural setting complicate a simple east-flowing versus west-flowing classification? Answer in 250 words.

STATIC THEORY

- **Claim:** Relief, water source, structure and sediment regime produce a broad Himalayan-Peninsular contrast, but exceptions make a direction-only classification inadequate.
- **Himalayan evidence/example:** Indus, Ganga and Brahmaputra have high-relief headwaters; they are mostly perennial with rain, snow/ice and groundwater contributions that vary seasonally. Some transverse gorges are antecedent, but antecedence cannot be pasted onto every trunk.
- **Peninsular evidence/example:** The older plateau broadly tilts east from a near-western divide. Mahanadi, Godavari, Krishna and Cauvery are long east-flowing systems; their regime is monsoon-dominant, though major rivers/reaches may be perennial or regulated.
- **Structural qualification/example:** Narmada and Tapi traverse structurally controlled rift/linear troughs westward to the Gulf of Khambhat, while short Western-Ghat streams also flow west. Estuarine/deltaic mouth form depends on sediment and marine energy, not direction alone.
- **Verdict:** Compare both systems on relief, regime, network history, channel behaviour and mouth form, then use structural and seasonal exceptions to refine the general rule.
- **Why this earns marks:** It satisfies Compare with common axes, gives named evidence on both sides, avoids false absolutes and turns exceptions into analysis.

MAINS ANGLE

Use a two-column table and a small eastward-tilt/rift-trough sketch.

STUDY LINK

Sections 7-11; 2021, 2024 and 2026 map PYQs.

HIGH

Original 20-mark practice - water security under uncertainty

GS-I | Prelims
Geography

INTRO & ORIGIN

Question: Assess inter-basin transfer as a water-security strategy under climate uncertainty. Use the Ken-Betwa case, but do not treat project implementation as proof of outcome. Answer in 250 words.

STATIC THEORY

- **Claim:** Transfer can be a conditional water-security tool but should be authorised and operated through revisable seasonal basin accounts, not a fixed surplus narrative.
- **Named evidence/example:** NWDA's April 2026 Jal Vikas and current KBLP material identify Ken-Betwa as under implementation, diverting Ken-basin water toward the water-deficit Betwa basin. This proves a real project/stage and direction, not completed benefit delivery.
- **Hydrological analysis:** A donor test must include precipitation, evapotranspiration, soil/aquifer/storage change, existing withdrawals, downstream/e-flow/sediment needs and monsoon timing. A recipient test includes demand profile, conveyance, treatment, crop/hydrogeology fit and local alternatives.
- **Benefit analysis:** Storage/conveyance can buffer a specific shortage; flood moderation requires designated flood space/operations; navigation requires depth, locks, terminals and sediment management. These claims cannot be assumed from a canal alone.
- **Named trade-off/example:** Panna landscape/ILMP context, downstream-flow/sediment, livelihood/resettlement and MP-UP-Union coordination demonstrate environmental/social/federal conditions.
- **Qualification and complements:** Climate uncertainty changes means and extremes. Combine transfer where justified with watershed work, recharge, efficiency/crop alignment, reservoir operations, floodplain restoration, wastewater reuse and transparent basin institutions.
- **Verdict:** Judge delivery outcomes separately from construction, and use monitoring/revision triggers to prevent a transfer from externalising risk.
- **Why this earns marks:** It assesses with explicit criteria, makes precise dated use of the case, explains mechanisms and limits, gives alternatives, and concludes conditionally.

MAINS ANGLE

Draw Ken -> Betwa plus four gates: seasonal balance, e-flow/sediment, recipient demand and social/federal legitimacy.

STUDY LINK

2020 GS-I Q14; NWDA April 2026 source; Sections 13-16.

End of Register Notes - UPSC Agent / Copilot CLI